PART II: PATTERNS

Topic 15: Lagged Effects — Why Policies Don't Work Instantly

The Core Concept

A **lag** is a time delay between an action (such as a policy change or training program) and its visible effect in the data. In the real world, almost nothing takes effect immediately.

Why It Matters: Avoiding Premature Abandonment

A common mistake in public safety policy evaluation: a new de-escalation training program launches in January, numbers do not drop in February, and leadership declares the program a failure. But behavior change takes time — officers need to internalize new skills, supervisors need to reinforce them, and the effects need to accumulate across enough incidents to move the monthly average.

The Example

Note: All data in this tutorial series are synthetic, created for teaching purposes only. They do not represent real-world incidents or any specific jurisdiction.

Hypothetical scenario based on Precinct North data:

Month	UoF Count	Training Status	Interpretation
Jan	22	✅ Launched	Training begins — no behavioral change yet
Feb	24	In progress	Short-term numbers may even rise (increased reporting awareness)
Mar	21	In progress	Still within normal noise range
Apr	18	Reinforced	Early signs of change — or just noise?
May	16	Reinforced	Downward signal becoming visible
Jun	14	Embedded	Clear effect: rate declining despite higher activity
Jul	17	Embedded	Summer spike — but lower than prior year (25 vs. 35 in 2022)

In this scenario, judging the program a failure in February would be premature — the actual effect only becomes visible from May onward, roughly 4–5 months after launch.

💡 **The Insight:** Expect a "slow-turning ship" effect from any new policy — not a reset button. The larger the organization and the more embedded the old behavior, the longer the lag.

Key Takeaway: When evaluating a new strategy, define the expected lag in advance (e.g., "we expect to see results by Month 4–6"). This prevents premature cancellation of effective programs.